

Week 3 - Part 3 - Osprey1st College Algebra - Summer 2024

Section covered: 3.3

Because the square of a real number is never negative, there is no real number x such that $x^2 = -1$. To provide a setting in which such equations have solutions, mathematicians invented an expanded system of numbers, the complex numbers. The **imaginary number i** , defined to be a solution of the equation $x^2 = -1$, is the basis of this new set.

The Imaginary Unit i

The **imaginary unit i** is defined as

$$i = \sqrt{-1}, \text{ where } i^2 = -1.$$

$$\begin{aligned} i^2 &= \sqrt{-1}^2 \\ i^2 &= -1 \end{aligned}$$

$$\begin{aligned} \sqrt{-25} &= \sqrt{25} \sqrt{-1} \\ &= 5i \end{aligned}$$

Complex Numbers and Imaginary Numbers

The set of all numbers in the form

$$a + bi,$$

with real numbers a and b , and i , the imaginary unit, is called the set of **complex numbers**. The real number a is called the **real part** and the real number b is called the **imaginary part** of the complex number $a + bi$.

Examples of complex numbers:

$5 + 2i$ - real part 5 , imaginary part 2 ; nonreal complex number

$1 - 3i$ - real part 1 , imaginary part -3 ; nonreal complex number

$4i = 0 + 4i$ - real part 0 ; imaginary part 4 ; nonreal complex number called also pure imaginary number

$3 = 3 + 0i$ - real part 3 , imaginary part 0 ; real number.

Note: Real numbers are special cases of complex numbers, where imaginary part is equal to zero!

Square Root of a Negative Number

Consider the following roots:

$$\sqrt{-16} = \sqrt{(-1)16} = \sqrt{-1}\sqrt{16} = i \cdot 4 = 4i;$$

$$\sqrt{-25} = \sqrt{(-1)25} = \sqrt{-1}\sqrt{25} = i \cdot 5 = 5i$$

The Expression $\sqrt{-a}$

If a is a positive real number then

$$\sqrt{-a} = i\sqrt{a}$$

Problem 8. Write as the product of a real number and i , using the definition of $\sqrt{-a}$.

(a) $\sqrt{-81} = 9i$

(b) $\sqrt{-98}$

$7i\sqrt{2}$

$\begin{array}{c} 2 \quad 49 \\ \nearrow \quad \nearrow \\ 7 \quad 7 \end{array}$

Adding and Subtracting Complex Numbers

Adding and Subtracting Complex Numbers

$$(a + bi) + (c + di) = (a + c) + (b + d)i \quad \text{and} \quad (a + bi) - (c + di) = (a - c) + (b - d)i$$

Note: Treat real parts as like terms and imaginary parts as like-terms.

Problem 9. Add or subtract two complex numbers.

(a) $*(2 + 5i) + (3 - 2i)$

$5 + 3i$

(b) $*(-10 + 9i) - (-13 + 10i)$

$+13 - 10i$

$3 - i$

$$\boxed{a+bi}$$

Problem 11. Divide two complex numbers.

(a) $\frac{(-2+6i)i}{i} = \frac{-2i+6i^2}{i^2} = -2i+6(-1) = \frac{-6-2i}{-1} = \boxed{6+2i}$

(b) $\frac{(5-2i)i}{-10i} = \frac{5i-2i^2}{-10i} = \frac{5i-2(-1)}{-10(-1)} = \frac{2+5i}{10} = \frac{2}{10} + \frac{5i}{10} = \boxed{\frac{1}{5} + \frac{1}{2}i}$

(c) FOIL $\frac{(3+4i)(2-i)}{(2+i)(2-i)}$
 $\frac{6-3i+8i-4i^2}{4+1-i^2} = \frac{6-3i+8i-4(-1)}{4+1-(-1)} = \frac{6-3i+8i+4}{4+1+1} = \frac{10+5i}{6}$

(d) FOIL $\frac{(1-5i)(-2+3i)}{(-2-3i)(-2+3i)}$
 $\frac{-2+3i+10i-15i^2}{4+9-i^2} = \frac{-2+3i+10i-15(-1)}{4+9-(-1)} = \frac{-2+3i+10i+15}{4+9+1} = \frac{13+13i}{14}$

Problem 12. * Perform the indicated operations.

(a) $\sqrt{-16} \cdot \sqrt{-25}$
 $4i \cdot 5i = 20i^2 = 20(-1) = \boxed{-20}$

(b) $\sqrt{-14} \cdot \sqrt{7}$
 $i\sqrt{14} \cdot \sqrt{7} = i\sqrt{98} = i\sqrt{49 \cdot 2} = 7i\sqrt{2} = \boxed{7i\sqrt{2}}$

(c) $\frac{\sqrt{4}}{\sqrt{-9}}$
 $\frac{2}{3i} \cdot \frac{3i}{3i} = \frac{6i}{9i^2} = \frac{6i}{9(-1)} = \boxed{-\frac{2}{3}i}$

(d) $\frac{\sqrt{-40}}{\sqrt{-2}}$
 $\frac{2i\sqrt{10}}{i\sqrt{2}} = \boxed{2\sqrt{5}}$

$$(e) (\sqrt{-7} + 3) \cdot (2i) = (i\sqrt{7} + 3)(2i) = \underbrace{2i^2\sqrt{7} + 6i}_{-2\sqrt{7} + 6i}$$

$$(f) \frac{1 + \sqrt{-9}}{2 - \sqrt{-4}} \quad \begin{array}{l} \text{FOIL} \\ a+bi \end{array} \frac{(1 + 3i)(2 + 2i)}{(2 - 2i)(2 + 2i)}$$

$$\frac{2 + 2i + 6i + 6i^2}{2 + 8i - 6}$$

$$\frac{-4 + 8i}{8} = \frac{-1 + 2i}{2} = \boxed{-\frac{1}{2} + i}$$

$4 + 4 = 8$

$\div \text{ by } 4$

Powers of i

#	0	$i^0 = 1$	$i^0 = 1$	
0.25	1	$i^1 = i$	$i^1 = i$	$i^5 = i^4 \cdot i = 1 \cdot i = i$
0.5	2	$i^2 = -1$	$i^2 = -1$	$i^6 = i^4 \cdot i^2 = 1 \cdot (-1) = -1$
0.75	3	$i^3 = -i$	$i^3 = i^2 \cdot i = (-1) \cdot i = -i$	$i^7 = i^4 \cdot i^3 = 1 \cdot (-i) = -i$
			$i^4 = i^2 \cdot i^2 = (-1)(-1) = 1$	$i^8 = i^4 \cdot i^4 = 1 \cdot 1 = 1$ and so on.

Powers of i cycle through the same four outcomes (i , -1 , $-i$, and 1) because i^4 has the same multiplicative property as 1 . Also, any power of i with an exponent that is a multiple of 4 has value 1. As with real numbers, $i^0 = 1$.

Problem 13. Simplify each power of i .

$$(a) i^{80} \quad (i^2)^{40} = (-1)^{40} = \boxed{1}$$

$$\frac{80}{4} = 20 \text{ r } 0 \quad i^0 = \boxed{1}$$

$$(b) i^{14} \quad \frac{14}{4} = 3 \text{ r } 2 \quad i^2 = \boxed{-1}$$

$$(c) i^{33} \quad \frac{33}{4} = 8 \text{ r } 1 \quad i^1 = \boxed{i}$$

8.25

$$(d) i^{15} \quad (i^2)^7 i = (-1)^7 i = \boxed{-i}$$

$$\frac{15}{4} = 3 \text{ r } 3 \quad i^3 = \boxed{-i}$$

$$(e) i^{-43} \quad \frac{43}{4} = 10 \text{ r } 3 \quad i^3 = (i^2)i = -i = -i$$

$$\frac{1}{i^{43}} = \frac{1}{-i} = \frac{i}{-i^2} = \frac{i}{-(-1)} = \frac{i}{1} = \boxed{i}$$

$$(f) i^{-36} \quad \frac{36}{4} = 9 \text{ r } 0 \quad i^0 = 1$$

$$\frac{1}{i^{36}} = \frac{1}{1} = \boxed{1}$$